CLR-DLR: A Semi-Supervised Framework for High-Fidelity Remote Sensing Segmentation

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***Abstract*—Remote sensing images are crucial for large-scale earth observation, providing vital data for land use, urban development, and environmental monitoring. However, the complexity of semantic segmentation tasks and the need for extensive pixel-level annotations pose significant challenges. To overcome these obstacles, researchers have explored semi-supervised approaches, consistency regularization, pseudo-labeling, and generative adversarial network (GAN)-based methods. Despite their potential, these methods often struggle with issues like oversmoothing, noisy label propagation, and training instability. To address this problem, this article introduces a novel semi-supervised model that combines pseudo-labeling with consistency regularization to address multiscale challenges and domain gaps. By implementing contextual label realignment (CLR) in the label space, we refine noisy labels, while denoising latent restoration (DLR) in the feature space uses a conditional diffusion model to tackle latent uncertainty. Our experimental results show that this framework effectively manages diverse imaging conditions and blurred boundaries, achieving superior robustness and accuracy with limited annotations. In addition, a polygonization-based boundary refinement enhances segmentation quality for complex object shapes, ensuring coherent delineation. In addition, the full code for reproducing all experiments will be available at https://github.com/A-leyenda/CLR-DLR**

***Index Terms*—Diffusion model, label refinement, remote sensing, semi-supervised segmentation, smart sensing.**

I. Introduction

REMOTE sensing images are essential for large-scale earth observation, offering critical insights into land use, urban development, and environmental monitoring. As these tasks become more complex, the importance of semantic segmentation—identifying and categorizing various objects in these images has grown significantly. However, developing high-performance segmentation models typically requires extensive pixel-level or region-level annotations, which are labor-intensive and time-consuming to produce.

In recent years, to mitigate data deficiencies, some studies have focused on improving datasets and classification strategies for enhanced remote sensing image processing [1]. Other work leverages the zero-shot learning capability of the CLIP model to achieve more efficient data pruning and sampling in large-scale scenarios [2]. Meanwhile, additional research has proposed multiscale skip connection networks enhanced with atrous convolutions to address segmentation challenges in multimodal and multiscale high-resolution remote sensing images [3], as well as to perform spatiotemporal fusion by leveraging multiscale mechanisms and feature concatenation—thus achieving superior performance on multiple benchmark datasets [4]. Building on these efforts, researchers have introduced semi-supervised learning algorithms into remote sensing semantic segmentation tasks, which can be broadly categorized into three representative approaches: consistency regularization-based methods [5], [6], pseudo-labeling-based methods [7], [8] [9], and generative adversarial network (GAN)-based methods [10].

While these methods have shown promise, several significant challenges remain. One primary obstacle lies in the multiscale nature of remote sensing data, where variations in spatial resolutions and the presence of both large- and small-scale objects can compromise the robustness of existing approaches. Another challenge arises from the diversity of imaging conditions and scene variations, which can lead to substantial domain gaps. For example, a model trained on data from a particular geographic area may struggle when tested on images from different regions or under varying imaging conditions. Moreover, blurred boundaries of certain targets, such as buildings or roads, further complicate the segmentation process. Addressing these complexities highlights the need for more specialized semi-supervised learning strategies tailored to remote sensing applications.

Specifically, for consistency regularization-based methods, though effective at leveraging unlabeled data via augmentation, they often yield over-smoothed predictions or struggle with complex domain variations if the augmentations fail to capture critical remote sensing scenarios. Meanwhile, pseudo-labeling-based methods, while expanding the labeled set with initial model predictions, can propagate noisy labels and thus accumulate errors across training iterations without explicit correction strategies. Last, GAN-based methods, which rely on adversarial training to discriminate real versus generated

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segmentations, may encounter training instability or mode collapse, especially given the high-resolution complexity of remote sensing imagery.

To address these multifaceted challenges in remote sensing segmentation—ranging from multiscale variability and domain gaps to blurred boundaries—we propose a semi-supervised framework that couples pseudo-labeling with consistency regularization, complemented by two specialized modules in different spaces: contextual label realignment (CLR) for refining labels and denoising latent restoration (DLR) for stabilizing latent representations. The system first uses a U-Net to produce multiple variants of pseudo-labels, including confidence-based versions, weakly augmented versions, and one derived from a direct segmentation output. These pseudo-labels are then passed through a semantic context embedding process that examines their relationships and refines them accordingly. Meanwhile, latent features extracted by the model's encoder are fed into a separate denoising component, which suppresses noise and yields more reliable internal representations. By simultaneously addressing label noise and feature-level uncertainty, this approach achieves stable pseudo-label generation and delivers robust performance even under limited annotation scenarios. Finally, a polygonization-based boundary refinement stage converts pixel-level predictions into vector polygons, sharpening object contours and further mitigating the impact of blurred boundaries.

Our research can be summarized by the following primary contributions.

1) *Multiscale Feature Extraction:* We integrate multiscale feature extraction directly into the training phase, enabling the model to learn both large- and small-scale object representations under limited annotations. This targeted approach effectively addresses scale variability in remote sensing data and is a key driver of our framework's performance gains.
2) *Label Noise Mitigation and Latent Denoising:* We apply strong consistency regularization on unlabeled data to address domain gaps stemming from varied imaging conditions. Beyond that, our framework introduces CLR, which significantly reduces label noise by aligning pseudo-labels to ground-truth-like contexts, effectively improving cross-region generalization. In parallel, the proposed DLR extends a diffusion-based denoising strategy to the model's deeper layers, ensuring that feature representations remain robust despite limited or partially erroneous labels. Together, CLR and DLR tackle both label and feature-level uncertainties, thus delivering markedly improved stability and segmentation accuracy under domain shifts.
3) *Polygonization-Based Boundary Refinement:* We employ a polygonization-based correction strategy that transforms coarse pixel-level segmentations into cleaner and shape-consistent polygons. By adjusting boundary vertices and smoothing contours, this approach effectively remedies segmentation noise and domain-related inconsistencies, resulting in sharper object edges—especially crucial when dealing with blurred targets such as roads or buildings.

II. Related Work

A. Semi-Supervised Semantic Segmentation in Remote Sensing

In the early stages of remote sensing image analysis, many approaches relied on manually crafted low-level features (e.g., color, texture, and shape) and traditional machine learning models for land-cover classification [11]. However, these "feature-engineering" methods often struggle to handle the complex, multiscale nature of remote sensing data. With the emergence of deep learning, fully supervised networks such as U-Net [12], DeepLab [13], and SegFormer [14], leveraging multiscale receptive fields and robust representation capabilities, have achieved significant advances in remote sensing semantic segmentation, albeit at the cost of large-scale, high-quality labeled datasets that are expensive and time-consuming to obtain.

In contrast, unsupervised methods such as SAM [15] and MemoryAdaptNet [16] require no labels but often suffer from interclass confusion and a lack of semantic priors, making them less effective in complex remote sensing scenarios. While semi-supervised learning strikes a balance by using a small amount of labeled data for initial guidance, then tapping into large-scale unlabeled images through consistency regularization or pseudo-labeling [17], [18]. This allows the model to maintain (or closely approximate) fully supervised performance while reducing annotation demands. For example, in urban building extraction, annotating a few key regions can provide prior knowledge, enabling pseudo-label generation and iterative refinement on wider unlabeled areas. This "small-to-large, iterative-refinement" paradigm offers an efficient and scalable pathway for remote sensing semantic segmentation across diverse regions and multisensor settings.

B. Latent Space Refinement and Diffusion Models

While many semi-supervised methods focus on output-level consistency, recent works highlight the importance of refining representations at deeper network layers [19]. By imposing consistency within the latent space, models become more resilient to annotation scarcity and noisy labels. In parallel, diffusion-based models have shown impressive results in generative tasks and image restoration [20], prompting researchers to explore them for iterative noise-removal in segmentation [21], [22]. However, leveraging conditional diffusion to specifically enhance latent feature alignment and robustness in remote sensing segmentation remains an underexplored direction.

C. EfficientNet in Segmentation Frameworks

Traditional backbones (e.g., ResNet [23] and VGG [24]) remain popular in segmentation pipelines, but EfficientNet [25] stands out by jointly scaling depth, width, and resolution for improved performance with fewer parameters. Recent studies suggest that integrating EfficientNet as an encoder in U-Net-like designs can yield more discriminative multi-level features [26]. Nonetheless, coupling EfficientNet with advanced semi-supervised modules, such as label realignment

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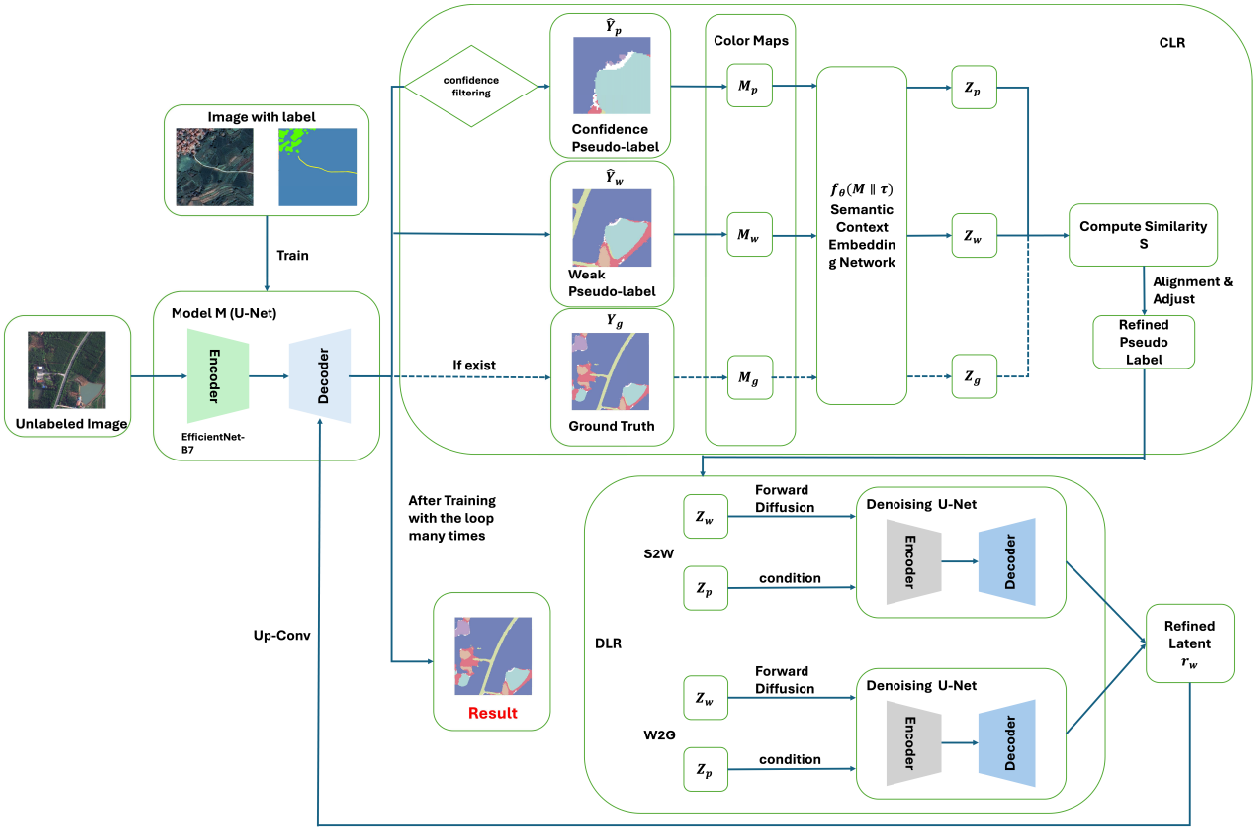


Fig. 1. Architecture of the overall model.

and diffusion-based latent refinement, has not yet been comprehensively investigated in remote sensing. We position our work to fill this gap by combining a high-capacity EfficientNet backbone with robust label and latent consistency mechanisms, thus offering a promising approach to large-scale Earth observation tasks.

III. Methodology

In this section, we detail our proposed semi-supervised segmentation framework, which integrates CLR and DLR into a U-Net architecture equipped with an EfficientNet-based encoder. Fig. 1 illustrates the high-level pipeline, from backbone feature extraction to refinement modules.

A. Overview

Our method aims to alleviate reliance on exhaustive annotations by leveraging unlabeled data through robust label refinement and latent-space consistency. Concretely, we adopt the following.

1) *EfficientNet-B7 as Encoder*: capturing multiscale hierarchical features from remote sensing images.
2) *CLR*: aligning pseudo-labels against ground-truth context to reduce label noise.
3) *DLR*: employing a conditional denoising diffusion model to restore deep feature consistency.

The decoder follows a standard U-Net structure with skip connections from the encoder. Semi-supervised learning is achieved by combining supervised losses on labeled data and auxiliary constraints on unlabeled samples (e.g., pseudo-labeling and diffusion-based latent constraints).

B. EfficientNet-Based U-Net

Instead of a traditional U-Net encoder (e.g., stacked convolutions), we adopt an EfficientNet-B7 [25] backbone for stronger feature extraction. Given an input image $\mathbf{I} \in R^{3\times H\times W}$, we obtain multiscale features $\{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$ from five stages of the EfficientNet. Each stage refines spatial resolution while increasing the receptive field, culminating in a 1/32 scale at $\mathbf{x}_4$ with 2560 channels for B7.

1) Decoder: We then upsample the deepest representation $\mathbf{x}_4$ in a symmetric U-Net decoder. Let $\mathbf{u}_4$ denote the initial decoder feature (e.g., projected from $\mathbf{x}_4$). For $i = 3, 2, 1, 0$, we merge the skip-connected encoder feature $\mathbf{x}_i$

$$\mathbf{u}_i = \text{UpBlock}\left(\mathbf{u}_{i+1}\right) \oplus \mathbf{x}_i \tag{1}$$

where $\oplus$ represents concatenation along the channel dimension, and UpBlock$(\cdot)$ typically consists of bilinear upsampling (or transpose convolution) followed by convolutions and nonlinear activations. Finally, $\mathbf{u}_0$ is passed through a 1×1 convolution to produce the segmentation logits $\hat{\mathbf{Y}}$ for C classes.

C. Contextual Label Realignment

In our pipeline (see Fig. 1), each unlabeled image produces two pseudo-labels, $\hat{\mathbf{Y}}_p$ (e.g., confidence-based) and $\hat{\mathbf{Y}}_w$

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(weakly augmented), whereas each labeled image additionally provides ground truth $\mathbf{Y}_g$. Although these pseudo-labels can guide semi-supervised training, they frequently contain misclassifications. We thus propose a CLR module to capture interclass relationships and refine noisy labels by exploiting color-based encodings and label-quality calibration.

1) Color-Based Label Encoding: Let $\hat{\mathbf{Y}}_p, \hat{\mathbf{Y}}_w, \mathbf{Y}_g \in R^{H\times W\times C}$ denote one-hot representations of up to C classes (for unlabeled images, $\mathbf{Y}_g$ does not exist). We assign each category a distinct RGB color and map each one-hot label to an $H \times W \times 3$ color image. Concretely, define a set of C RGB tuples $\{\mathbf{c}_1, \ldots, \mathbf{c}_C\}$, each in $[0, 255]$, and for each pixel (h, w), if its class index is k, we look up the corresponding color $\mathbf{c}_k$. Hence, we obtain colored label maps $\mathbf{M}_p, \mathbf{M}_w, \mathbf{M}_g \in R^{H\times W\times 3}$, where $\mathbf{M}_g$ is only used if the ground truth is available. This color encoding provides visually interpretable cues and enhances class separability prior to embedding.

2) Embedding and Calibration: Next, we embed these colored maps into latent features via a lightweight CNN block $f_\theta(\cdot)$. For unlabeled data, we focus on $\mathbf{M}_p, \mathbf{M}_w$, optionally incorporating a scalar τ^u to indicate their agreement

$$\tau^u = \text{Dice}\left(\hat{\mathbf{Y}}_p, \hat{\mathbf{Y}}_w\right). \tag{2}$$

For labeled data, we can define

$$\tau^l = \text{Dice}\left(\hat{\mathbf{Y}}_w, \mathbf{Y}_g\right) \tag{3}$$

to gauge how closely the weak pseudo-label aligns with the ground truth. A larger τ value implies greater consistency (i.e., confidence). The embedding thus proceeds as

$$\begin{aligned} \{\mathbf{z}_p, \mathbf{z}_w\} &= f_\theta\left(\mathbf{M}_p, \mathbf{M}_w \parallel \tau^u\right) \quad \text{(unlabeled)} \\ \{\mathbf{z}_w, \mathbf{z}_g\} &= f_\theta\left(\mathbf{M}_w, \mathbf{M}_g \parallel \tau^l\right) \quad \text{(labeled)} \end{aligned} \tag{4}$$

yielding latent features $\mathbf{z}_p, \mathbf{z}_w, \mathbf{z}_g \in R^{d\times h\times w}$.

3) Contextual Similarity and Realignment: We then compute a pixel-wise similarity map between two relevant embeddings, e.g., $\mathbf{S} = \text{Sim}(\mathbf{z}_p, \mathbf{z}_w)$ for unlabeled samples, or $\text{Sim}(\mathbf{z}_p, \mathbf{z}_g)$ if we wish to compare pseudo-labels against ground truth. Each $\mathbf{S}(h, w)$ reflects how closely the features align at position (h, w). Low $\mathbf{S}(h, w)$ suggests higher uncertainty or potential misclassification in $\hat{\mathbf{Y}}_p$.

A realignment function $\mathcal{A}$ then refines the original pseudo-label

$$\widetilde{\mathbf{Y}}_p = \mathcal{A}\left(\hat{\mathbf{Y}}_p, \mathbf{S}\right) \tag{5}$$

where low-similarity pixels may be replaced by $\hat{\mathbf{Y}}_w$ or smoothed out in ambiguous regions, while confident predictions remain intact. We likewise produce a refined label $\widetilde{\mathbf{Y}}_w$ if comparing $\mathbf{z}_w$ with $\mathbf{z}_g$, etc. This step reduces noise and enhances semantic consistency, leading to better supervision for unlabeled images.

4) Overall CLR Workflow: Summarizing, the CLR module proceeds through four main stages in a sequential manner: First, it performs **Color Encoding** by mapping one-hot labels $\{\hat{\mathbf{Y}}_p, \hat{\mathbf{Y}}_w, \mathbf{Y}_g\}$ into color images $\{\mathbf{M}_p, \mathbf{M}_w, \mathbf{M}_g\}$, ensuring each category is assigned a distinct hue. Next, in the **Embedding with Calibration** step, these color maps (plus a scalar τ^u or τ^l) are passed into f_θ to obtain latent features $\{\mathbf{z}_p, \mathbf{z}_w, \mathbf{z}_g\}$. Subsequently, the **Similarity Computation** phase compares pairs of embeddings (e.g., $\mathbf{z}_p$ versus $\mathbf{z}_w$) to form a spatial similarity map $\mathbf{S}$. Finally, in the **Realignment** stage, any noisy pseudo-labels are adjusted according to $\mathbf{S}$, producing refined labels $\widetilde{\mathbf{Y}}_p$ and $\widetilde{\mathbf{Y}}_w$ with reduced noise and improved consistency.

By integrating color-based embeddings and a dice-driven calibration factor, CLR effectively mitigates errors in self-generated pseudo-labels. Moreover, its context-aware realignment stage captures interclass relationships, thereby enhancing the supervision quality on unlabeled data and improving overall segmentation performance.

D. Denoising Latent Restoration

Although CLR focuses on label refinements, deeper semantic features can still suffer from limited annotations and noisy pseudo-labels. We hence propose DLR, which leverages a conditional denoising diffusion probabilistic model (cDDPM) on the encoder's bottom features to achieve robust latent-space rectification.

1) Motivation and Overview: While CLR refines pseudo-labels at the output space, the internal representations at deeper layers (e.g., the final-stage encoder feature) may remain suboptimal, especially when labeled data are scarce. To alleviate such a distribution mismatch, our DLR module treats feature embeddings as "noisy samples" and progressively refines them via a diffusion model. Concretely, for each unlabeled sample $\mathbf{I}_u$, we obtain two latent embeddings $(\mathbf{z}_s, \mathbf{z}_w)$ from the CLR module (e.g., "strong" versus "weak" pseudo-labels). For labeled samples $(\mathbf{I}_l, \mathbf{Y}_l)$, we similarly obtain $(\mathbf{z}_w, \mathbf{z}_l)$ for "weak" pseudo-label and ground truth, respectively. In both cases, DLR learns to denoise (or transport) the weaker embedding toward a cleaner representation, conditioned on the stronger counterpart.

2) Forward Diffusion Process: Let $\mathbf{x}_{\text{start}}$ denote the "clean" reference feature to which we wish to align (e.g., the strong pseudo-label embedding or a ground-truth embedding), and $\mathbf{x}_t$ be the progressively noised version of $\mathbf{x}_{\text{start}}$ at timestep $t \in \{1, \ldots, T\}$. The forward diffusion process is

$$\mathbf{x}_t = \sqrt{\alpha_t}\,\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_t, \quad t = 1, \ldots, T \tag{6}$$

where $0 < \alpha_t < 1$ is a noise schedule and $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$. After T steps, $\mathbf{x}_T$ becomes nearly Gaussian.

A. Strong-to-weak transportation (S2W): For unlabeled data $\mathbf{I}_u$, we treat $\mathbf{z}_s$ (strong) as the cleaner reference feature $\mathbf{x}_{\text{start}}$, and $\mathbf{z}_w$ (weak) as the one we wish to denoise. We diffuse $\mathbf{z}_w$ forward to $\mathbf{z}_w^T$ and later learn to reverse this process conditioned on $\mathbf{z}_s$.

B. Weak-to-ground-truth transportation (W2G): For labeled data $\mathbf{I}_l$, we consider $\mathbf{z}_l$ (ground-truth) as the cleaner reference and $\mathbf{z}_w$ (weak pseudo-label) as the noisy input. The forward diffusion takes $\mathbf{z}_l$ to $\mathbf{z}_l^T$, and the model learns to reconstruct $\mathbf{z}_l$ with $\mathbf{z}_w$ as condition.

3) Reverse (Denoising) Process: A denoising U-Net $\boldsymbol{\epsilon}_\theta$ inverts the forward diffusion. For each step $t \to t-1$, it predicts the distribution of $\mathbf{x}_{t-1}$ given $\mathbf{x}_t$ and a condition (e.g., $\mathbf{z}_s$ or $\mathbf{z}_l$)

$$\begin{aligned} p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{z}_{\text{cond}}, t) = \mathcal{N}\big(&\mathbf{x}_{t-1} \mid \boldsymbol{\mu}_\theta(\mathbf{x}_t, \mathbf{z}_{\text{cond}}, t) \\ &\boldsymbol{\sigma}_\theta^2(\mathbf{x}_t, \mathbf{z}_{\text{cond}}, t)\big). \end{aligned} \tag{7}$$

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TABLE I

DATASET OVERVIEW AND SEMI-SUPERVISED SPLIT (EXAMPLE FOR L = 20%)

Dataset	Training		Validation	
	Labeled	Unlabeled	Labeled	Unlabeled
LoveDA	504	2,018	334	1,335
iSAID	6,796	27,182	2,329	9,315

TABLE II

COMPARISON OF PARAMETERS AND FPS WITH SOTA

Model	Parameters (M)	FPS
HRNet [29]	–	16.74
DeepLabV3+ [13]	–	75.33
DCSwin[30]	50.0	–
SegFormer [31]	13.7	31.30
UNetFormer [32]	11.7	115.60
Ours	**68.0**	**22.22**

TABLE III

VALIDATION METRICS WITH 20% LABEL USED ON LOVEDA AND ISAID

Dataset	mIoU	Dice	F1	Recall	Precision	Accuracy
LoveDA	57.22	69.96	69.96	68.21	73.45	73.41
iSAID	69.56	70.26	81.55	81.74	82.24	82.42

A. Training objectives: For unlabeled (S2W), we diffuse $\mathbf{z}_w$ to $\mathbf{z}_w^T$ and train

$$\mathcal{L}_{\text{lat–u}} = E_{\mathbf{z}_w,\mathbf{z}_s,t} \left\| \boldsymbol{\epsilon}_\theta \left(\mathbf{z}_w^t, \mathbf{z}_s, t\right) - \boldsymbol{\epsilon} \right\|^2 \tag{8}$$

and similarly for labeled (W2G)

$$\mathcal{L}_{\text{lat–l}} = E_{\mathbf{z}_l,\mathbf{z}_w,t} \left\| \boldsymbol{\epsilon}_\theta \left(\mathbf{z}_l^t, \mathbf{z}_w, t\right) - \boldsymbol{\epsilon} \right\|^2 . \tag{9}$$

4) Feature Reconstruction and Fusion: After training, given a weak feature $\mathbf{z}_w$, we can sample random noise $\mathbf{r}_w^T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively apply the reverse steps conditioned on $\mathbf{z}_s$ to obtain a rectified $\mathbf{r}_w$. We then fuse $\mathbf{r}_w$ with the encoder output

$$\mathbf{x}_{\text{ref}} = \text{ConvBlock}\left([\mathbf{x}_4,\ \mathbf{r}_w]\right) \tag{10}$$

making the final decoded features more robust under limited supervision.

A. Summary of DLR: Overall, DLR introduces a latent-space diffusion mechanism to rectify noisy feature embeddings. By learning to transport weak features to stronger or ground-truth References, it effectively boosts representation quality. Coupled with label-level CLR refinement, our method upholds consistency and robustness in both output-label space and deep feature space.

E. Polygonization-Based Boundary Refinement

While DLR improves the latent representations, remote sensing imagery often exhibits blurred or irregular boundaries. Hence, after generating the final segmentation masks, we apply a lightweight polygonization step to trace object outlines and convert them into vector polygons. We then perform simple vertex smoothing, effectively sharpening object contours for large man-made structures such as roads and buildings, without incurring significant additional computational cost.

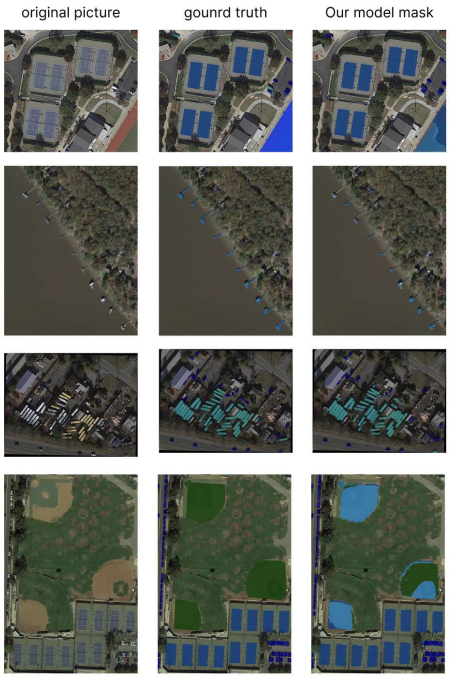


Fig. 2. Visual performance of our model on iSAID.

F. Semi-Supervised Training Procedure

Let $\mathcal{D}_L$ be the labeled set and $\mathcal{D}_U$ the unlabeled set. Each mini-batch contains labeled examples $(\mathbf{I}^l, \mathbf{Y}^l)$ and unlabeled ones $\mathbf{I}^u$.

1) Supervised Loss: For $(\mathbf{I}^l, \mathbf{Y}^l)$, we employ a weighted sum of cross-entropy $\mathcal{L}_{\text{CE}}$ and Dice loss $\mathcal{L}_{\text{Dice}}$

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{CE}}\left(\hat{\mathbf{Y}}^l, \mathbf{Y}^l\right) + \mathcal{L}_{\text{Dice}}\left(\hat{\mathbf{Y}}^l, \mathbf{Y}^l\right). \tag{11}$$

2) CLR-Based Pseudo-Labels: For unlabeled $\mathbf{I}^u$, the model predicts initial $\hat{\mathbf{Y}}^u$, which is further refined by CLR to yield $\widetilde{\mathbf{Y}}^u$. This refined label supervises the unlabeled image in a consistency loss

$$\mathcal{L}_{\text{unsup}} = \text{MSE}\left(\hat{\mathbf{Y}}^u,\ \widetilde{\mathbf{Y}}^u\right) + \text{optional constraints} \tag{12}$$

ensuring predictions match refined pseudo-labels under data augmentations.

3) DLR Latent Consistency: We also incorporate $\mathcal{L}_{\text{lat}}$ from (13), which enforces robust latent representation reconstruction

$$\begin{aligned}\mathcal{L}_{\text{lat}} = {} & \lambda_u\, E_{(\mathbf{z}_s,\mathbf{z}_w)\sim\mathcal{D}_U,\, t\sim[1,T]} \left\| \boldsymbol{\epsilon}_\theta \left(\mathbf{z}_w^t,\, \mathbf{z}_s,\, t\right) - \boldsymbol{\epsilon} \right\|^2 \\ & + \lambda_l\, E_{(\mathbf{z}_l,\mathbf{z}_w)\sim\mathcal{D}_L,\, t\sim[1,T]} \left\| \boldsymbol{\epsilon}_\theta \left(\mathbf{z}_l^t,\, \mathbf{z}_w,\, t\right) - \boldsymbol{\epsilon} \right\|^2 .\end{aligned} \tag{13}$$

4) Overall Objective: Combining these yields our training objective

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_{\text{unsup}}\, \mathcal{L}_{\text{unsup}} + \lambda_{\text{lat}}\, \mathcal{L}_{\text{lat}}. \tag{14}$$

IV. EXPERIMENTS AND RESULTS

In this section, we present a comprehensive evaluation of our semi-supervised segmentation framework, which is built upon an EfficientNet-based U-Net backbone, incorporates strong data augmentation, and utilizes both consistency and latent-refinement losses (via cDDPM). We provide details on

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TABLE IV
ABLATION STUDY ON LOVEDA

Model Variant	mIoU (%)
Base U-Net	43.74
+EfficientNet-B7 + CLR module	54.20
+DLR module	55.10
+Polygonization Boundary Refinement (**Full pipeline**)	**57.22**

TABLE V
ABLATION STUDY ON ISAID

Model Variant	mIoU (%)
Base U-Net	56.08
+EfficientNet-B7 + CLR module	66.52
+DLR module	67.48
+Polygonization Boundary Refinement (**Full pipeline**)	**69.56**

TABLE VI
ABLATION STUDY WITH VARYING PROPORTIONS OF LABELED DATA ON LOVEDA

Label Usage	mIoU	Dice	F1	Recall	Precision	Accuracy
1%	49.81	60.72	60.72	57.80	4.0	68.15
5%	53.49	67.66	67.66	65.24	70.94	70.27
10%	55.47	66.94	66.94	64.61	72.19	72.66
20%	57.22	69.96	69.96	68.21	73.45	73.41

TABLE VII
ABLATION STUDY WITH VARYING PROPORTIONS OF LABELED DATA ON ISAID

Label Usage	mIoU	Dice	F1	Recall	Precision	Accuracy
1%	62.35	63.20	74.30	77.40	71.50	73.50
5%	65.83	66.65	78.58	81.46	76.52	77.46
10%	66.75	67.44	79.66	81.67	78.64	78.84
20%	69.56	7.026	81.55	81.74	82.24	82.42

the experimental setup, implementation, evaluation metrics, comparative results, and ablation studies. The overall training procedure follows the code described in Section III, particularly leveraging ramp-up schedules, polynomial LR, and test-time augmentation (TTA).

A. Dataset and Semi-Supervised Splits

We evaluate our method on two large-scale remote sensing benchmarks, LoveDA [27] and iSAID [28], under the same limited-annotation setup. LoveDA provides 2522 high-resolution images (0.3–1.0 m/pixel) for training and 1669 for validation, each annotated with seven land-cover categories (background, building, road, water, barren, forest, and agricultural). The iSAID dataset contains 33 978 aerial images initially annotated for 15 instance-level categories, which we convert into semantic masks by merging instance labels. Following its official release, we use 22 334 images for training and 11 644 for validation. In both datasets, only $L\%$ of the training images are sampled as labeled, leaving $(100 - L)\%$ unlabeled. For example, when $L = 20\%$ in LoveDA, we select 504 labeled images out of 2522, with the remaining 2018 unlabeled; similarly, in iSAID, $L = 20\%$ corresponds to labeling 4466 images out of 22 334. We then apply the same semi-supervised process, generating pseudo-labels for unlabeled images, refining them with our proposed modules, and conducting final evaluations on the respective validation splits, while repeating the random sampling multiple times to reduce bias in any single labeled/unlabeled partition.

TABLE VIII
COMPARISON OF MIOU METHODS ON ISAID DATASET

Method	Label Usage	mIoU (%)
PSPNet [33]	100%	60.3
DeepLabV3 [13]	100%	59.0
HRNet [29]	100%	62.3
FarSeg [34]	100%	63.7
PFNet [35]	100%	66.9
Segformer [31]	100%	65.6
FactSeg [36]	100%	64.8
BSNet [37]	100%	63.4
RSP-Swin-T [38]	100%	64.1
Ringmo [39]	100%	67.2
RSSFormer [40]	100%	65.9
AerialFormer-B [41]	100%	69.3
Hierarchical Class Graph Fusion [42]	100%	65.21
RemoteSAM-FT [43]	100%	67.01
UperNet(InterImage-XL+UniRS) [44]	100%	68.00
MA3E(ViT-B) [45]	100%	64.06
Ours	**20%**	**69.6**

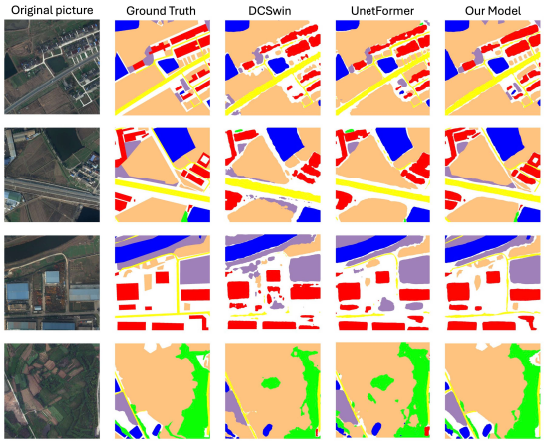


Fig. 3. Comparison of visual results with other SOTA methods on LoveDA.

For those unlabeled images, during training, we first apply the current model to each unlabeled image to generate initial pseudo-labels. These pseudo-labels are then refined by the CLR module, which embeds them into a semantic color space and adjusts misaligned regions via a similarity-based function. Concurrently, we pass the encoder's bottom-layer feature to the DLR module, which uses conditional diffusion to produce a more robust latent representation. Finally, the refined pseudo-labels supervise the network through a consistency loss under various augmentations, promoting stable predictions on unlabeled data.

Table I summarizes the total images, labeled/unlabeled splits, and validation sets for both datasets under an example 20% labeled scenario. In practice, we vary $L\% \in \{1, 5, 10, 20\}$.

B. Experimental Setup

Our codebase is implemented in PyTorch and trained on two NVIDIA RTX 4090 GPUs. We adopt an AdamW optimizer with an initial learning rate of $1e^{-4}$ (decayed using a polynomial schedule). Following a semi-supervised paradigm, we combine a supervised loss (cross-entropy + dice) for

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TABLE IX
COMPARISON OF IOU PER LABEL (%) ON LOVEDA DATASET

Model	Label Usage	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU
U-Net (ResNet50) [27]	100%	43.06	52.74	52.78	73.08	10.33	43.05	59.87	47.84
HRNet (W32) [27]	100%	44.61	55.34	57.42	73.96	11.07	45.25	60.88	49.79
DeepLabv3+ (ResNet50) [27]	100%	42.97	50.88	52.02	74.36	10.04	44.21	58.53	47.62
AerialFormer-BF [41]	100%	47.8	60.79	59.93	81.5	17.29	49.04	60.94	54.1
SegFormer-B5 [31]	100%	46.54	57.46	59.81	82.79	20.89	46.14	61.0	53.49
DC-Swin small [30]	100%	45.9	57.69	58.72	83.81	18.57	46.63	59.86	53.85
UPerNet (Swin small) [46]	100%	46.18	59.97	57.41	81.79	20.93	46.66	61.11	53.24
UPerNet (ViT-C12x4) [47]	100%	47.57	61.6	59.91	81.79	18.26	48.00	62.64	54.4
UNetFormer (ResNet18) [32]	100%	44.7	58.8	58.04	82.1	23.54	47.88	60.84	53.99
MTP (InternImage-XL) [48]	100%	46.8	62.6	58.96	82.25	17.49	47.39	63.34	54.17
AerialFormer [41]	100%	47.8	60.7	59.3	81.5	17.9	47.9	64.0	54.1
SAIP-Net [49]	100%	55.36	64.53	57.66	71.77	28.90	43.53	59.44	54.46
LightFormer [50]	100%	45.9	59.3	54.2	81.5	27.8	47.7	63.7	54.3
ST-DASegNet (SegFormer[51])	100%	36.78	59.83	43.77	73.83	19.3	49.96	67.01	50.08
U-Net Ensemble [52]	100%	49.09	61.12	63.71	82.36	30.15	49.29	65.82	57.36
Ours	**20%**	**43.15**	**68.41**	**68.82**	**80.32**	**24.32**	**57.68**	**57.86**	**57.22**

labeled samples with an unsupervised consistency loss and a latent consistency loss derived from our diffusion module. Batch size is set to 12, with a labeled batch size of 6 when mixing labeled and unlabeled data in each iteration. We ramp up unsupervised loss weights linearly over the first few thousand iterations to stabilize early training. Unless otherwise stated, each experiment is run for 50 000 iterations, typically completing within 12 h on the dual-GPU setup.

C. Model Complexity and Timing

Our final network, combining an EfficientNet-B7 backbone, the CLR module, and the DLR module, contains ~68 million parameters.[1] When evaluated with an input resolution of 512 × 512, the model requires about ~145 GFLOPs per forward pass. On a single NVIDIA RTX 4090, we measure an average of ~45 ms per image in inference mode (batch size = 1), including the overhead of our final polygonization step. Hence, while somewhat heavier than older CNN baselines, the framework remains practical for large-scale remote sensing applications where accuracy is paramount.

To further quantify runtime efficiency, we convert latency into frames per second (FPS) using the formula

$$\text{FPS} = \frac{1000}{\text{Inference Time (ms)}}$$

which yields $(1000/45) \approx 22.22$ FPS for our model. We compare our model's complexity and FPS against several existing semantic segmentation baselines in Table II.

D. Evaluation Method

We measure segmentation performance using mean intersection over union (mIoU) and class-wise IoU, along with other indicators such as dice, $F1$, precision, recall, and overall pixel accuracy. These metrics collectively capture how well predicted regions align with ground truth and how effectively positive samples are identified. In addition, we employ a TTA strategy—flipping and rotating images at inference—to mitigate sensitivity to spatial transformations. By aggregating the outputs across all augmented views, we obtain more stable and accurate final predictions.

[1]Exact parameter counts may vary slightly depending on specific PyTorch and CUDA versions.

E. Experimental Result

Table III lists the metrics achieved by our method on two different datasets, LoveDA and iSAID, using only 20% of the labeled data. On LoveDA, the model attains an IoU of 57.22 and a dice/$F1$ score of 69.96, coupled with a recall of 68.21 and a precision of 73.45, leading to an overall accuracy of 73.41. When applied to iSAID under the same label-scarce setting, the approach yields an mIoU of 69.56 and a dice/$F1$ score of 70.26 ($F1$ up to 81.55), along with favorable recall (81.74) and precision (82.24), giving an overall accuracy of 82.42. These outcomes illustrate the method's adaptability across distinct remote sensing scenarios without relying on large volumes of annotated data.

Fig. 2 further demonstrates our model's segmentation performance by presenting several representative examples on iSAID. In each composite image, the original remote sensing tile is shown on the left, the ground-truth mask in the middle, and our model segmentation output on the right.

F. Ablation Studies

To isolate the contributions of each component, we conduct ablation experiments on key modules as follows.

1) Base U-Net without any additional unsupervised losses.
2) Base U-Net + EfficientNet-B7 + CLR module.
3) Base U-Net + EfficientNet-B7 + CLR module + DLR module.
4) Full Model (Base U-Net + EfficientNet-B7 + CLR module + DLR module + Polygonization Boundary Refinement).

Tables IV and V show that either consistency loss or latent refinement alone brings notable improvements over the base model. Combining both yields the highest mIoU, indicating

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TABLE X

COMPARISON RESULTS WITH DIFFERENT LABEL RATIOS ON THE LOVEDA DATASET. BOLD INDICATES THE BEST RESULT IN EACH COLUMN WITHIN THE SAME RATIO BLOCK. IOU, MIOU, AND $F1$ ARE GIVEN AS PERCENTAGES

Label Usage	Model	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU(%)	F1
1.00%	CCVC [53]	34.15	9.43	10.55	14.90	1.09	24.97	11.97	15.29	25.24
	CutMix [54]	36.04	24.69	10.03	24.60	3.43	6.67	10.19	16.52	26.85
	CCT [55]	37.16	22.41	27.86	43.98	14.51	25.38	36.67	29.71	44.99
	CPS [56]	46.52	20.87	27.85	50.55	0.01	33.16	34.60	30.51	44.28
	OnlySup	41.96	37.86	36.74	57.03	9.02	31.70	37.73	36.01	51.41
	Mean teacher [57]	44.73	42.53	40.34	50.92	11.37	26.88	54.40	38.73	52.81
	DebiasPL [58]	34.85	**47.78**	44.05	55.60	21.09	33.06	52.63	41.29	57.50
	U2PL [59]	46.58	46.30	50.29	64.75	20.07	29.55	39.59	42.45	58.31
	LSST [18]	44.73	41.90	39.90	62.65	29.27	31.26	48.29	42.57	59.00
	UniMatch [60]	46.53	51.38	49.36	67.74	10.86	33.40	52.28	44.51	59.51
	FixMatch [61]	46.78	51.20	50.21	67.27	11.53	**36.79**	50.26	44.86	60.02
	AEL [62]	44.92	49.28	48.69	64.86	28.71	35.40	47.36	45.60	61.84
	DWL [63]	48.74	56.79	51.59	63.42	22.56	35.20	55.38	47.67	63.40
	AllSpark [64]	63.87	47.70	46.05	61.52	35.31	30.94	55.64	48.72	63.29
	MUCA [65]	**64.89**	56.03	47.14	63.86	**35.81**	22.57	**58.18**	49.78	**63.48**
	Ours(CLR-DLR)	39.02	58.27	**59.20**	**69.58**	19.46	48.88	49.25	**49.81**	60.72
5.00%	CutMix [54]	41.48	41.62	38.77	47.44	14.69	28.09	31.05	34.73	50.65
	CCVC [53]	44.17	42.82	35.09	51.20	3.17	31.40	44.62	36.07	50.93
	CPS [56]	48.90	49.64	47.97	60.27	4.67	36.09	47.32	42.12	56.90
	CCT [55]	46.80	44.62	46.80	60.95	24.83	29.03	44.30	42.48	58.74
	OnlySup	48.00	43.34	50.56	61.42	23.14	37.63	45.00	44.20	60.46
	Mean teacher [57]	49.73	46.22	42.34	60.93	31.51	35.79	44.22	44.39	61.81
	FixMatch [61]	45.40	53.05	51.22	66.73	28.53	27.25	54.30	44.64	62.45
	U2PL [59]	52.58	53.12	50.97	65.75	16.48	38.16	47.89	46.42	61.93
	AEL [62]	45.40	53.05	51.22	66.73	28.53	27.25	54.30	46.64	62.45
	UniMatch [60]	50.20	54.49	50.46	67.18	26.79	30.06	54.86	47.72	63.46
	LSST [18]	51.48	45.66	52.66	67.63	33.52	35.80	48.60	47.91	64.10
	DebiasPL [58]	46.83	56.21	53.12	55.70	31.72	38.42	53.57	47.94	64.22
	DWL [63]	48.75	55.00	51.53	69.49	29.46	36.59	52.11	48.99	64.88
	AllSpark [64]	65.09	55.06	47.59	67.10	34.67	26.86	51.87	49.75	64.91
	MUCA [65]	**67.29**	56.04	48.37	61.02	**36.21**	30.76	**57.09**	50.97	64.92
	Ours(CLR-DLR)	40.34	**63.95**	**64.32**	**75.06**	22.73	**53.96**	54.14	**53.49**	**67.66**
10.00%	CCVC [53]	46.79	40.25	46.96	45.79	19.57	26.58	38.38	37.76	54.01
	CCT [55]	44.07	45.22	47.65	57.12	24.41	32.50	45.07	42.29	58.73
	CutMix [54]	46.73	49.60	47.36	59.99	29.06	37.77	40.60	44.44	60.99
	CPS [56]	51.30	54.93	52.57	53.37	18.39	37.59	53.24	45.91	61.78
	OnlySup	47.02	51.04	50.73	59.42	31.66	38.25	51.26	47.05	63.52
	DebiasPL [58]	45.35	57.13	51.56	55.64	32.08	36.79	54.74	47.61	63.97
	Mean teacher [57]	50.45	55.75	43.56	66.15	35.24	36.96	45.64	47.68	64.18
	UniMatch [60]	51.80	53.95	51.17	58.15	25.60	38.72	54.86	47.75	63.86
	U2PL [59]	51.44	53.44	53.43	56.82	29.44	39.55	51.89	48.00	64.32
	LSST [18]	50.69	49.50	52.63	69.85	27.25	36.24	52.06	48.32	64.17
	AEL [62]	49.69	53.84	52.58	69.55	27.92	39.14	46.84	48.51	64.35
	FixMatch [61]	52.02	55.59	53.20	57.91	25.86	40.83	**57.50**	48.99	64.97
	AllSpark [64]	67.13	56.16	40.67	63.58	32.54	32.03	56.91	49.86	63.97
	DWL [63]	49.94	56.66	53.89	70.35	30.62	41.49	53.13	50.87	66.64
	MUCA [65]	**68.69**	58.20	41.82	65.62	**37.09**	35.01	57.38	51.97	66.72
	Ours(CLR-DLR)	41.85	**66.35**	**66.75**	**77.86**	23.57	**55.92**	56.10	**55.47**	**66.94**

that consistency regularization at the output level and latent-level refinement are complementary for robust segmentation.

In addition, we further investigate how the proportion of labeled data impacts performance by training our models on only 5%, 10%, and 20% of the labeled dataset. Tables VI and VII report the results, showing that our approach continues to achieve strong performance even under highly label-scarce conditions, and consistently improves as more labeled data becomes available.

G. Compared With SOTA Models

For the result on the iSAID dataset, as shown in Table VIII, most existing methods rely on 100% labeled data yet still remain below our performance. In contrast, our approach achieves 69.6% mIoU using only 20% labeled data, surpassing previous CNN-based and transformer-based solutions in effectively segmenting remote sensing imagery.

As Table IX, on the LoveDA dataset, although our approach utilizes only 20% of the labeled data, it achieves higher class IoU scores in building, road, and forest than all methods using 100% labels. Furthermore, our method surpasses nearly all state-of-the-art architectures that rely on 100% supervision, including traditional CNN backbones like ResNet50, as well as Transformer-based networks such as SegFormer and AerialFormer, demonstrating a powerful modeling capability under limited supervision for capturing diverse land-cover features.

As shown in Table X, we also compare our method with recent state-of-the-art semi-supervised segmentation models at three different annotation ratios (1%, 5%, and 10%) on the LoveDA dataset. Our approach consistently outperforms competing methods, especially on challenging categories like building, road, and forest.

Moreover, for example, as illustrated in Fig. 3, a visual comparison against two of the SOTA methods, dcswin [30] and

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UNetformer [32], further highlights how our model produces more coherent segmentation boundaries and handles small-scale objects more accurately.

H. Limitations and Future Work

Despite achieving notable improvements on both the LoveDA and iSAID datasets, our method still faces the following challenges.

1) *Highly Scarce and Imbalanced Classes:* For classes that constitute less than 1% of the data or exhibit severe imbalance, pseudo-labels often remain noisy and biased toward majority classes, which hinders consistency-based training. Future work could explore rare-class over-sampling, class-adaptive weightings, or noise-aware denoising strategies specifically tailored for ultralow-frequency categories.
2) *Real-Time Inference Limitations:* Although our model achieves high accuracy, its inference speed on an RTX4090 is approximately 45 ms per 512×512 image (≈22.22 FPS), which falls short of strict real-time requirements for some applications (e.g., on-board processing). Optimizing architecture for latency, such as model pruning, quantization, or distillation, will be important for deployment in time-sensitive scenarios.
3) *Long-Range Context:* Extremely large geographic extents may benefit from architectural enhancements that can better capture long-range dependencies and global context in remote sensing imagery.
4) *Generalization Across Diverse Sensors:* While LoveDA spans urban/rural scenes and iSAID focuses on high-resolution aerial objects, extending the model to fundamentally different sensor data could require advanced domain adaptation to handle distinct spectral and spatial characteristics.

V. Conclusion

In this article, we propose a novel semi-supervised segmentation framework that couples a powerful EfficientNet-based U-Net backbone with two specialized modules: CLR and DLR. By refining pseudo-labels against more realistic semantic cues and enforcing latent-space consistency through a conditional diffusion mechanism, our approach effectively addresses the major challenges in semi-supervised remote sensing segmentation—including noisy pseudo-labels, domain gaps, and limited annotated data.

Extensive experiments on both the LoveDA and iSAID datasets demonstrate that our model consistently surpasses almost all other state-of-the-art methods, but with only 20% labeled data. Specifically, our multiscale feature extraction design proves crucial in tackling objects of varying sizes, while the polygonization-based boundary refinement stage notably sharpens object contours, mitigating the impact of blurred edges common in aerial imagery. The combination of CLR and DLR further reinforces robustness by reducing label noise at the output level and stabilizing encoder features in deep layers.

Despite these advancements, limitations remain in handling extremely rare classes and achieving scalability to large-scale geographic contexts. Future work may thus explore more sophisticated sampling or label-denoising strategies for infrequent categories, integrate multisensor data (e.g., multispectral and SAR) to further broaden applicability, and consider hybrid Transformer-CNN backbones for enhanced global reasoning. Overall, this study underscores the value of jointly refining label- and latent-spaces under a semi-supervised paradigm, paving the way for more accurate and resource-efficient remote sensing segmentation across diverse domains.

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